GinCol Lab · Aula

Student worksheet

Artistic Drawing I · 1st year of Bachillerato · Learning to look · Class 1 of 5 · With Marina

Why does almost nobody draw what they see?

Class video: https://youtu.be/gdQXig7NNRE

The code leads to the video.

Name: _____ Date: _____

Before you start

1. If you want to, on a sheet of paper, draw a face from the front, from memory, inside a rectangle. With the whole head, without taking any care over it: about ten seconds. Keep it without showing it to anyone: it's just for you, and you'll come back to it in this class and at the end of topic 2. How high up the head do you think you've put the eyes: above halfway, at halfway or below?

2. **At the end of the class.** Go back to your answer. Were you right? What has happened to the forehead of your face?

Measure your first face

With a ruler, on your drawing. Always measure from the chin. The head, up to the top of the skull, not the top of the hair; the eyes, up to the inner corner of the eye (the corner next to the nose).

	First face (from memory)	Second face (with the three steps)
Height of the head, from the chin to the top of the skull (cm)		
Height of the eyes, from the chin to the inner corner of the eye (cm)		
Eyes ÷ head × 100 (%)		
Above 52%, between 47 and 52%, or below?		

If you only drew the face, without the skull, you can't do the calculation. Write that down too: it's data.

If you want to: measure yourself in the mirror

Stand in front of a mirror with a pencil and measure **on your reflection, with the pencil resting on the glass and never close to your eye**. Don't measure anyone with a pencil close to their face. If it feels uncomfortable, don't do it: the class gives you the answer.

1. Put the tip of the pencil on the chin of your reflection and your thumb level with the inner corner of the eye.

2. Without moving your thumb, slide the pencil up until the tip is at the inner corner of the eye.

3. Where does your thumb reach? Count up to where the skull would be, not the hairstyle.

What does the result tell you about where your eyes are?

While you watch the video

1. **The study.** Who did the experiment with the faces, when did they publish it and with how many people? What did they measure? Where did the people who drew from memory put the eyes, on average, and where are they in an adult head?

2. **The 20 cm head.** How many centimetres from the chin are the eyes in a real 20 cm head, and how many did people put them at from memory? What disappears when you raise them? Which explanation fits the data best, and why is it called a hypothesis?

3. **Light and the retina.** What reaches your eye when you look at an apple? What is the image formed on the retina like? Is that what you see?

4. **The blind spot.** Why is there a blind spot in each eye? Why do you never notice it? What is the other "dark area" that you don't see either?

5. **The debate.** Does the brain paint in what's missing in the blind spot, or does it simply not miss it? Give both positions, each with who defends it and with its reason.

6. **The sentence in the decree.** What does it mean that what you see "is not a photo of the world"? "What is perceived is nothing more than the brain's reconstruction of the information received." Which part has been tested, and which part is a question for philosophy?

7. **Thouless's plate.** What were the proportions of the oval that reached the eye, and how did his observer draw it? What happened when the observer copied an ellipse drawn on paper, seen from the front? What does that show? What is that effect called, and what is it useful for in everyday life?

8. **Tracing and copying.** Why does a tracing come out better than a copy? What are the four steps of drawing, and which one fails?

9. **Three false ideas.** What are they, and what takes each one apart?

10. Why does the curriculum say that drawing is a form of knowledge? What was measured in the seven experiments with drawn words? What is the difference between seeing and looking?

11. **Ruskin and Gombrich.** What did each one argue, and what was his best argument? On what point did Gombrich agree with Ruskin?

Fact, hypothesis or debate?

Mark **F** if it's a fact (the data say so), **E** if it's an explanation that is still a hypothesis, or **D** if it's a question that is still being debated.

1. From memory, people put the eyes, on average, at 57% of the height of the head.

2. We raise the eyes because we draw the head like a box, flat on top.

3. In the blind spot there are no cells that capture light.

4. The brain paints in what's missing in the blind spot.

5. Out of 129 people, including ten artists and art teachers, not one escaped shape constancy.

6. Copying a face turned upside down does not improve the drawing.

7. What is perceived is nothing more than the brain's reconstruction.

8. Drawing a word helps you remember it better than writing it.

Practice

1. **The plate.** Put a plate on the table and draw it from your chair, without measuring. Then take a photo of it from the same spot, with your phone at the height of your eyes. Measure the height and width of the oval in your drawing and in the photo. Is your oval rounder than the one in the photo?

2. Explain in your own words why a tracing comes out better for you than a copy.

3. Draw another face from memory, this time with the eyes at halfway, and put it next to the first one. Follow the three steps: first, the whole head, with its skull; then, mark halfway; and on that line, the eyes. Measure it with the table above.

For exercise 1:

	Height of the oval (cm)	**Width of the oval (cm)**	**Height ÷ width**
My drawing			
The photo			

Keep thinking

Who is right, Ruskin or Gombrich? Can you draw what is there, without what you know, or is every drawing already an interpretation? Give each one's best argument before saying what you think.

For the whole topic

A couple of pencils, an ordinary one (HB) and a soft one (2B or 4B), an eraser and some sheets of paper. If a class needs anything else, it will tell you. Your drawings are yours: you don't need to show them or upload them anywhere.

The data in the class come from Carbon and Wirth (2014), from the studies of head measurements cited in the teacher sheet, from Thouless (1931), from Cohen and Bennett (1997) and from the other studies in the teacher sheet; the quotations, from Ruskin (1857) and Gombrich (1960).

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